

Problem 1: Given $\mathbf{F} = [2, 2y + z, y + 1]$

(a) Show that $\mathbf{F}$ is conservative in two different ways

method 1

Show $\text{curl}(\vec{F}) = \vec{0}$

method 2

Find f such that $\vec{F} = \nabla f$

$$f_x = 2$$

$$f = 2x + g(y, z)$$

$$2y + z = f_y = 0 + g_y$$

$$g = y^2 + zy + h(z)$$

$$f = 2x + y^2 + zy + h(z)$$

$$y + 1 = f_z = y + h_z$$

$$h_z = 1$$

$$h = z + C$$

$$\text{So } f(x, y, z) = 2x + y^2 + zy + z$$

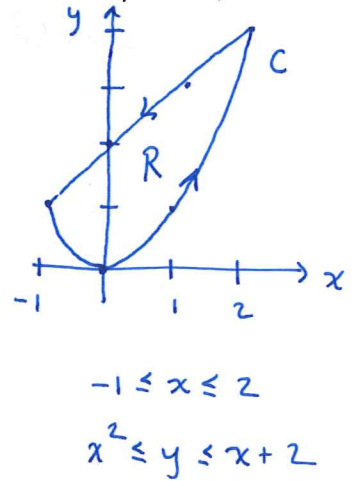
(b) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ when C is a straight line from $\overset{A}{(-1, 4, 2)}$ to $\overset{B}{(0, 0, -6)}$

$$\begin{aligned} &= \int_A^B \nabla f \cdot d\vec{r} = f(B) - f(A) \\ &= -6 - (-2 + 16 + 8 + 2) \\ &= -6 - 24 = -30 \end{aligned}$$

Problem 2: Let $F = [x + y, 2xy]$ and C be the boundary of the region bounded by $y = x^2$ and $y = x + 2$ in the x, y -plane.

(a) Compute $\oint_C F \cdot dr$ using green's theorem. $N_x = 2y$ $M_y = 1$

(b) Setup the integral you would use to compute $\oint_C F \cdot dr$ directly (don't leave answer as a dot product)



$$(a) \oint_C \vec{F} \cdot d\vec{r} = \iint_R (2y - 1) dA$$

$$= \int_{-1}^2 \int_{x^2}^{x+2} (2y - 1) dy dx$$

$$= \int_{-1}^2 \left[y^2 - y \right]_{x^2}^{x+2} dx$$

$$= \int_{-1}^2 \left[(x+2)^2 - (x+2) - x^4 + x^2 \right] dx$$

$$= \int_{-1}^2 \left[x^2 + 4x + 4 - x - 2 - x^4 + x^2 \right] dx = \int_{-1}^2 \left[-x^4 + 2x^2 + 3x + 2 \right] dx$$

$$= \left[-\frac{1}{5}x^5 + \frac{2}{3}x^3 + \frac{3}{2}x^2 + 2x \right]_{-1}^2$$

$$= -\frac{1}{5} + \frac{2}{3} + \frac{3}{2} + 2 - \left(-\frac{1}{5} - \frac{2}{3} + \frac{3}{2} - 2 \right) = -\frac{2}{5} + \frac{4}{3} + 4$$

(b) C consists of 2 curves : $y = x^2$ from $(-1, 1)$ to $(2, 4)$ and $y = x + 2$ from $(2, 4)$ to $(-1, 1)$

$$C_1: \vec{r}_1(t) = [t, t^2] \quad -1 \leq t \leq 2 \quad \text{and} \quad C_2: \vec{r}_2(t) = [t, t+2] \quad 2 \leq t \leq -1$$

$$\oint_C F \cdot dr = \oint_{C_1} \vec{F}(\vec{r}_1(t)) \cdot \vec{r}_1'(t) dt + \oint_{C_2} \vec{F}(\vec{r}_2(t)) \cdot \vec{r}_2'(t) dt$$

$$= \int_{-1}^2 [t + t^2, 2t^3] \cdot [1, 2t] dt + \int_2^{-1} [t + t + 2, 2t(t+2)] \cdot [1, 1] dt$$

$$= \int_{-1}^2 (t + t^2 + 4t^4) dt + \int_2^{-1} (2t + 2 + 2t^2 + 4t) dt$$

Problem 3: Let $\mathbf{F} = [x, y, z]$ and S be the sphere $x^2 + y^2 + z^2 = 25$. Compute $\iint_S \mathbf{F} \cdot \mathbf{n} dA$

S is closed, so let's apply the divergence Thm

$$\operatorname{div} \vec{F} = 1 + 1 + 1 = 3$$

let T be the solid sphere

$$\iint_S \vec{F} \cdot \vec{n} dA = \iiint_T 3 dV$$

$$= 3 \operatorname{Volume}(T)$$

$$= 3 \left(\frac{4}{3} \pi (5)^3 \right)$$

$$= 500\pi$$

Problem 4: Compute the flux of water (water density = $1 \frac{\text{kg}}{\text{m}^3}$) through the parabolic cylinder $S: y = x^2$ where $0 \leq x \leq 2$ and $0 \leq z \leq 1$ if the velocity flow of water is given by $\mathbf{v} = [3z^2, 6, 6xz]$ (units: m/s).

$$\rho = 1 \frac{\text{kg}}{\text{m}^3} \quad \vec{F} = \rho \vec{v} = [3z^2, 6, 6xz]$$

$$\text{Want} \quad \iint_S \vec{F} \cdot d\vec{A} = (*)$$

$$\begin{array}{ll} x = x & u = x \\ y = x^2 & v = z \\ z = z & \end{array}$$

$$\vec{r}(u,v) = [u, u^2, v] \quad \begin{array}{l} 0 \leq u \leq 2 \\ 0 \leq v \leq 1 \end{array}$$

$$\vec{r}_u = [1, 2u, 0]$$

$$F(r(u,v)) = [3v^2, 6, 6uv]$$

$$\vec{r}_v = [0, 0, 1]$$

$$\vec{r}_u \times \vec{r}_v = 2u \vec{i} - \vec{j} + 0 \vec{k} = [2u, -1, 0]$$

$$(*) \quad \iint_R [3v^2, 6, 6uv] \cdot [2u, -1, 0] \, du \, dv$$

$$= \int_0^1 \int_0^2 6uv^2 - 6 \, du \, dv$$

$$= \int_0^1 3u^2 v^2 - 6u \Big|_0^2 \, dv$$

$$= \int_0^1 12v^2 - 12 - 0 \, dv$$

$$= 4v^3 - 12v \Big|_0^1 = 4 - 12 = -8 \text{ kg/s}$$

Problem 5: Answer the following conceptual questions

- (a) Suppose $\mathbf{F} = [1, 1, 1]$. Is it true that $\iint_S \mathbf{F} \cdot \mathbf{n} dA = 0$ for any closed surface S which is a boundary of a solid? Justify your answer.

Since S is closed, we can apply the divergence theorem:

$$\operatorname{div} \vec{F} = 0$$

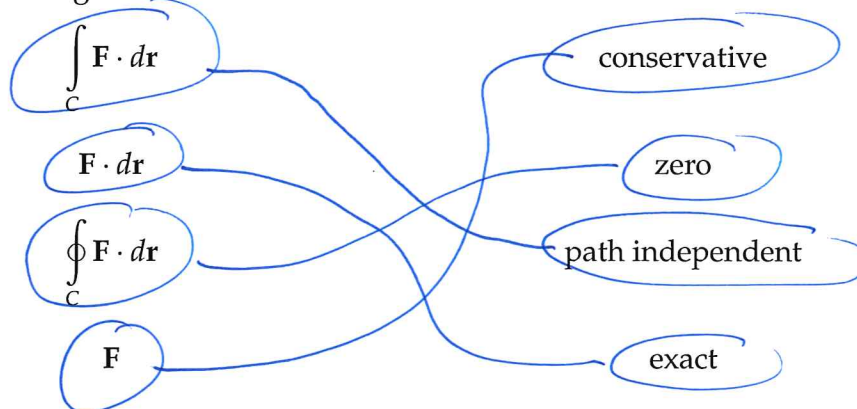
$$\Rightarrow \iint_S \vec{F} \cdot \vec{n} dA = \iiint_T \operatorname{div} F dV = 0$$

- (b) Describe the difference between Green's Theorem and Stokes' Theorem.

- (c) Briefly describe the meaning of flux.

Amount of some quantity (e.g. Mass) through a surface

- (d) Let $\mathbf{F}$ vector field such that $\operatorname{curl} \mathbf{F} = \mathbf{0}$. Match each expression in the left column with the most appropriate term in the right column.



- (e) Suppose we know to represent a hill as a surface given by the function $f(x, y)$. If we place a ball anywhere on the hill, what direction would you expect it to roll when we let it go?

$$-\nabla f(x, y)$$

